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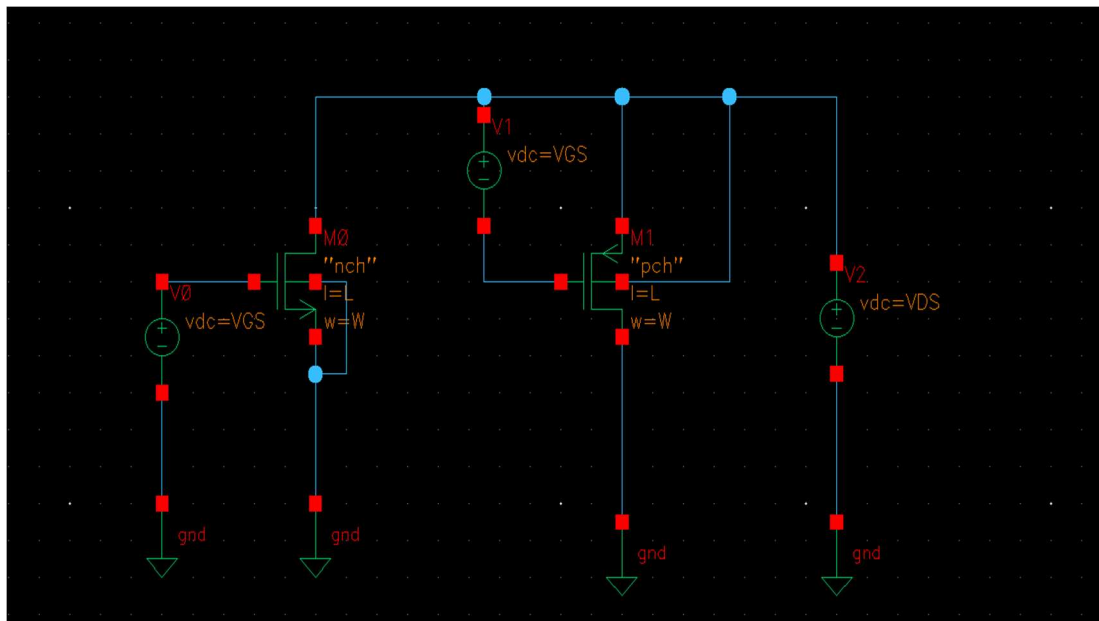
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Task 3

Part 1: Sizing Chart

DC Operating point Analysis

Schematic Test Bench



Given parameters

DC gain = -66

VDD=1.8v

Biasing current= 85uA

Random Number Generator

This version of the generator creates a random integer. | thousand digits.

Result

85

Lower Limit

Upper Limit

Generate  **Clear**

Biasing current

Random Number Generator

This version of the generator creates a random integer. | thousand digits.

Result

66

Lower Limit

Upper Limit

Generate  **Clear**

DC gain

$$A_v = -g_m r_o = -\frac{2I_D r_o}{V_{OV}}, \quad r_o = \frac{V_a}{I_D}$$

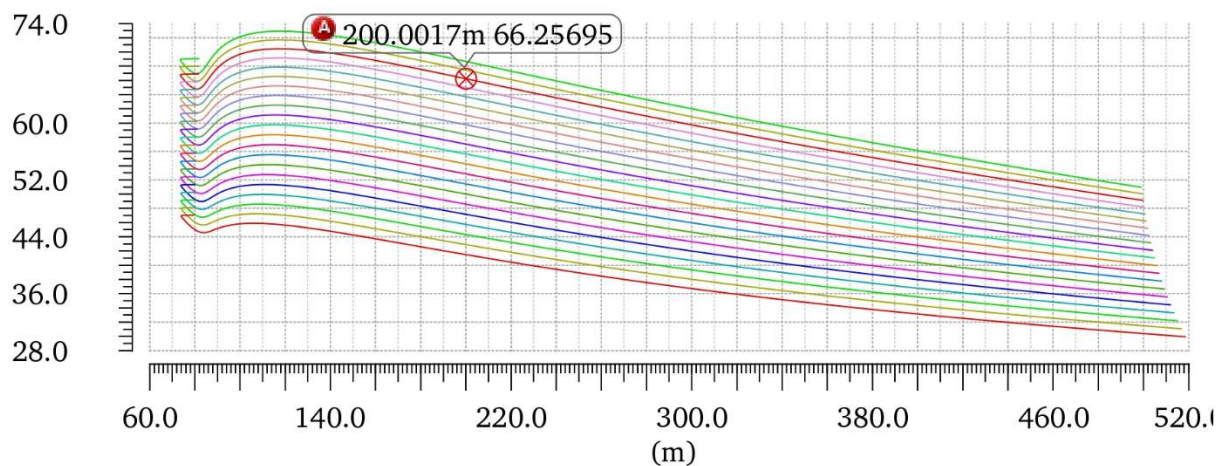
$$\text{Let } g_m = \frac{2I_D}{V^*}, \quad \text{where } V^* \approx V_{OV}$$

$$\text{So } |A_v| = \frac{2V_a}{V^*}$$

The L (channel length) is unknown and we want an L that should satisfy the value of two parameters first one is V^* and the second $g_m r_o$ so we will do a dc sweep with VGS nested in it a parametric sweep with L and from the family of curves of different Ls we will see which one at the V^* specified gives $g_m r_o$ to do this we used the wave vs wave function to plot the V^* at the x axis and $g_m r_o$ at Y-axis while leaving $w=10\mu\text{m}$

V^* given value =200mv

$g_m r_o = 66$



The point that meet our specs is at $L=290\text{nm}$ in which $V^* = 200\text{mv}$ & $g_m r_o = 66$



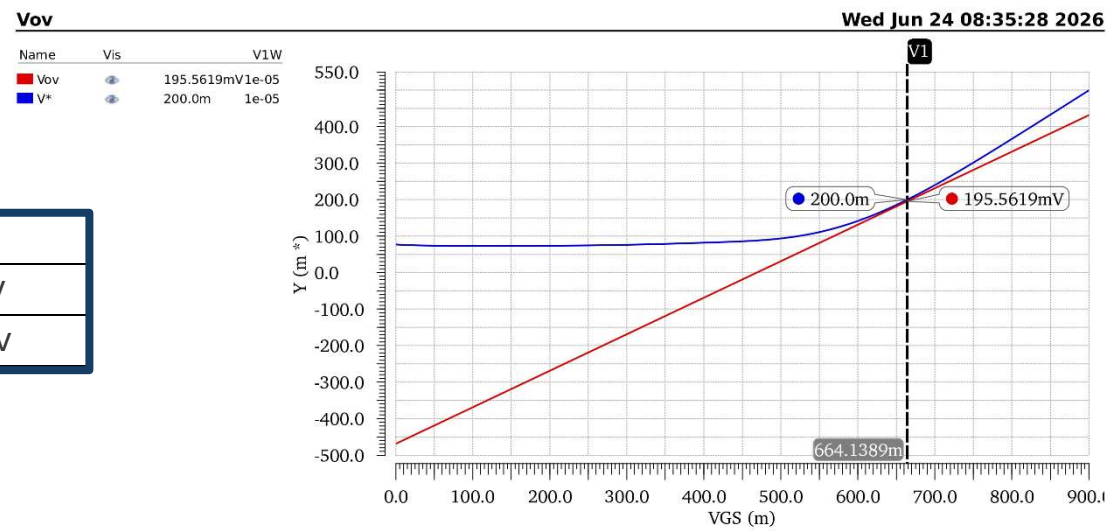
■	v* vs gmro	2.7e-07
■	v* vs gmro	2.75e-07
■	v* vs gmro	2.8e-07
■	v* vs gmro	2.85e-07
■	v* vs gmro	2.9e-07
■	v* vs gmro	2.95e-07
■	v* vs gmro	3e-07

Here is a clearer image of the parametric nested sweeps shows that $L = 290\text{nm}$ verifies what is needed

After we Have got the L we could proceed normally to determine the rest of the operating point parameters like V_{GSQ} and so we will do a DC sweep with V_{GS} and plot V^* and V_{ov} with V_{GS} while leaving The W with an arbitrary value of $10\mu\text{m}$ as we can get the needed value of W later through the bias current needed

V* & Vov vs VGS at W=10um

V*=200mv
Vov=195.5619mv
VGS=664.1389mv

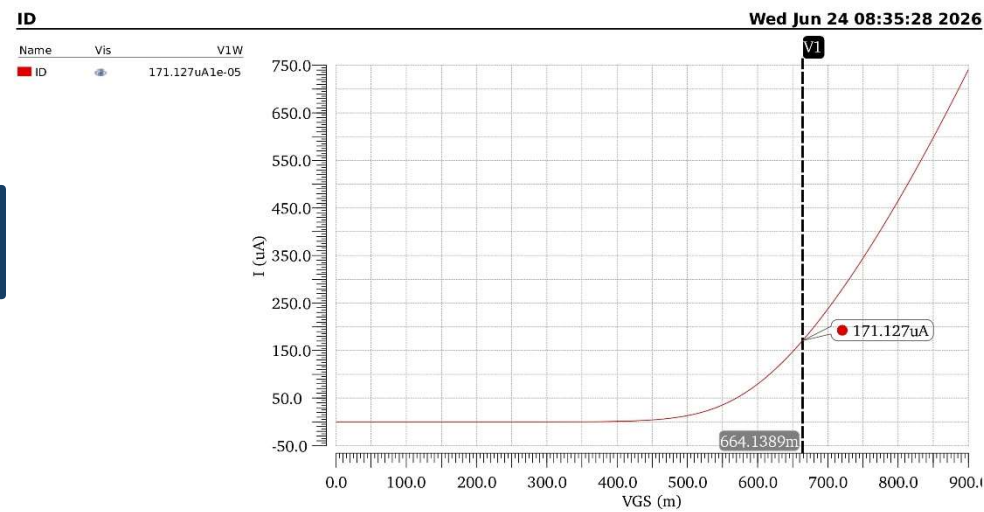


Markers

	W	V1
x		664.1389m
Vov		
Vov	1e-05	195.5619mV
V*		
V*	1e-05	200.0m

Idx vs VGS at W=10um

ID =171.127uA
VGS=664.1389mv



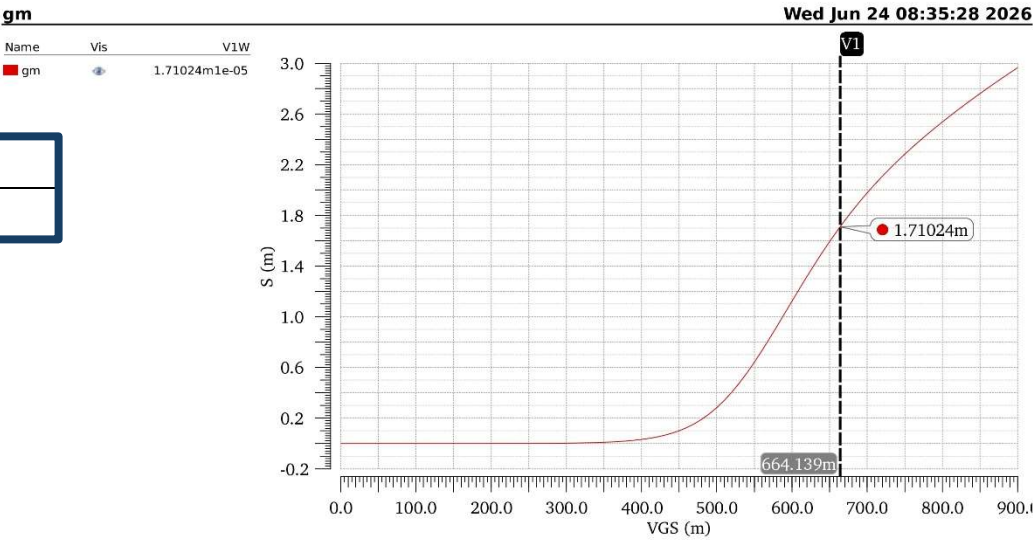
Markers

	W	V1
x		664.1389m
ID		
ID	1e-05	171.127uA

gm_x vs V_{GS} at W=10μm

gm = 1.71024mS

VGS=664.1389mv

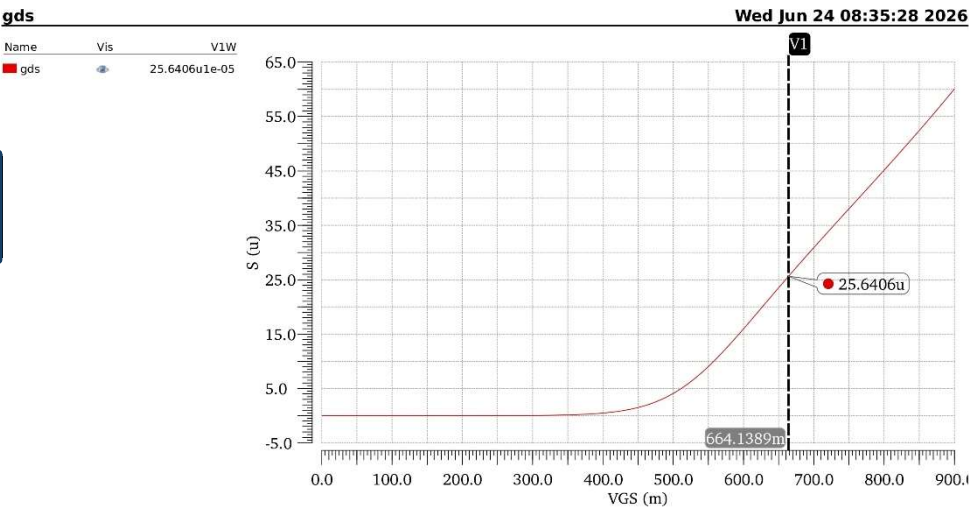


Markers		W	V1
	x		664.139m
	gm		
	gm	1e-05	1.71024m

gds_x vs V_{GS} at W=10μm

gds = 25.6uS

VGS=664.1389mv



Markers		W	V1
	x		664.1389m
	gds		
	gds	1e-05	25.6406u

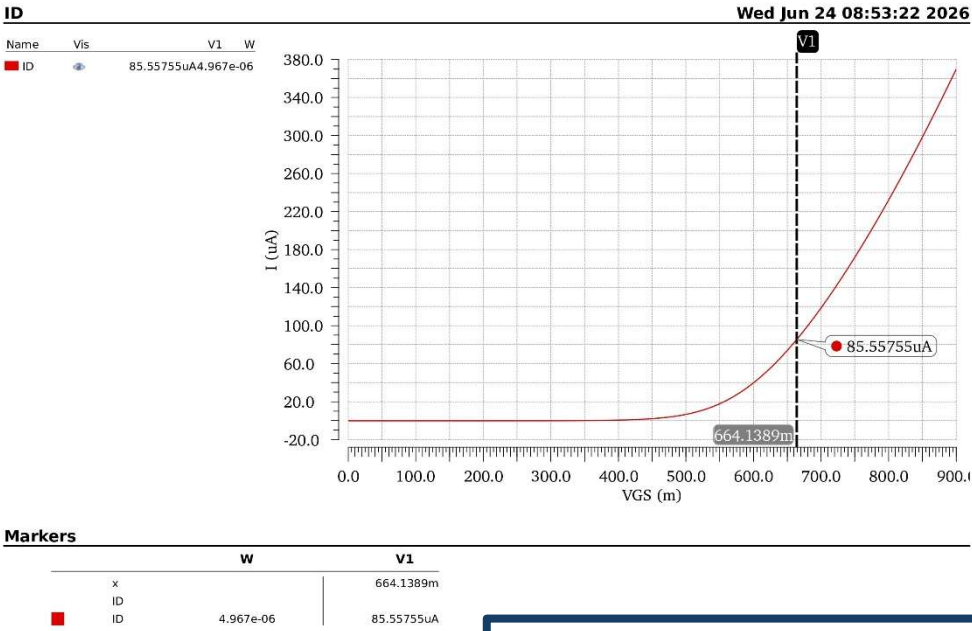
	At x point	At operating point
ID	171.127uA	85uA
W	10um	4.967um

	At x point	At operating point
gm	1.71024mS	849.47uS
W	10um	4.967um

	At x point	At operating point
gds	25.6uS	12.715uS
W	10um	4.967um

Idq vs VGS at W=4.967um

ID =85.55uA
VGS=664.1389mv



Here the current is a bit higher than the intended and this can be solved by a slight decrease in W and an in small increase L to compensate the intrinsic gain.

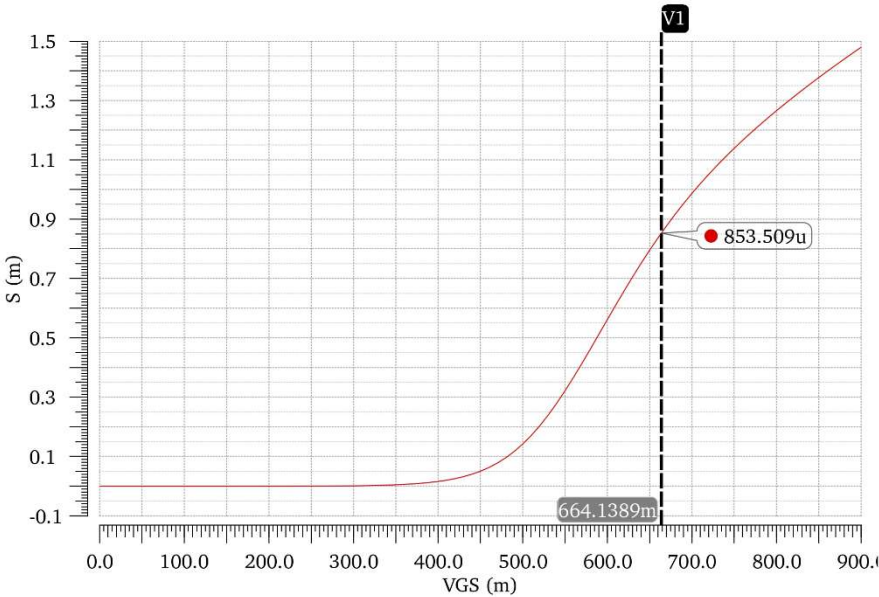
gmq vs VGS at W=4.967um

gm Wed Jun 24 08:53:22 2026

Name	Vis	V1W
gm		853.509u4.967e-06

gm =853.509uS

VGS=664.1389mv



Markers

	W	V1
x		664.1389m
gm		
gm	4.967e-06	853.509u

gdsq vs VGS at W=4.967um

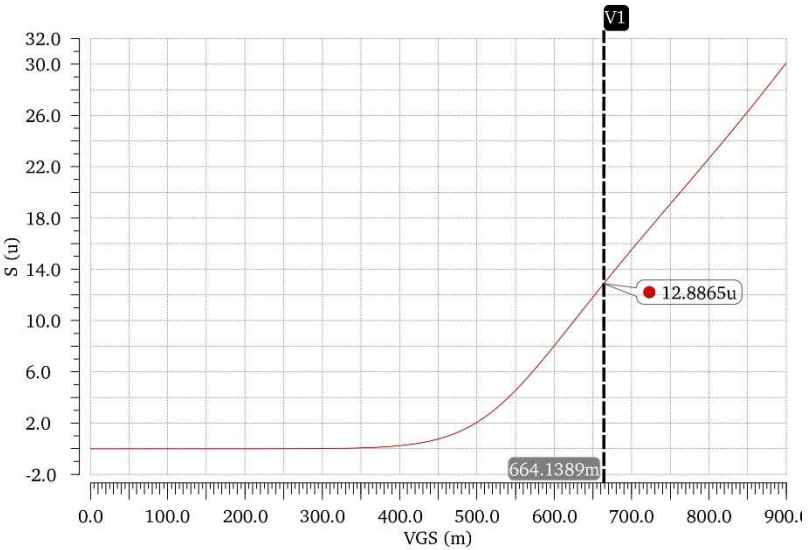
gds Wed Jun 24 08:53:22 2026

Name	Vis	V1W
gds		12.8865u4.967e-06

gds =12.8865uS

VGS=664.1389mv

$$A_v = -g_m r_o = -\frac{853.5}{12.8865} = -66.2$$



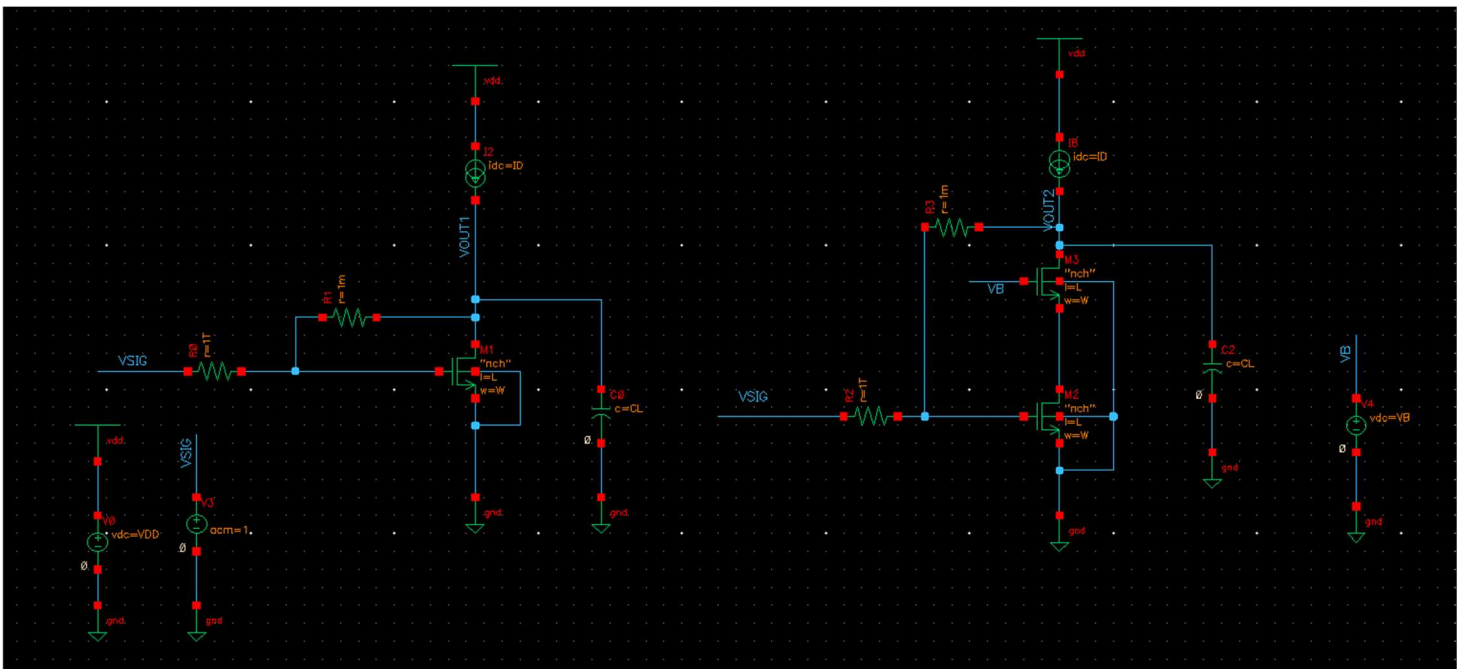
Markers

	W	V1
x		664.1389m
gds		
gds	4.967e-06	12.8865u

PART 2: Cascode for Gain

OP Analysis

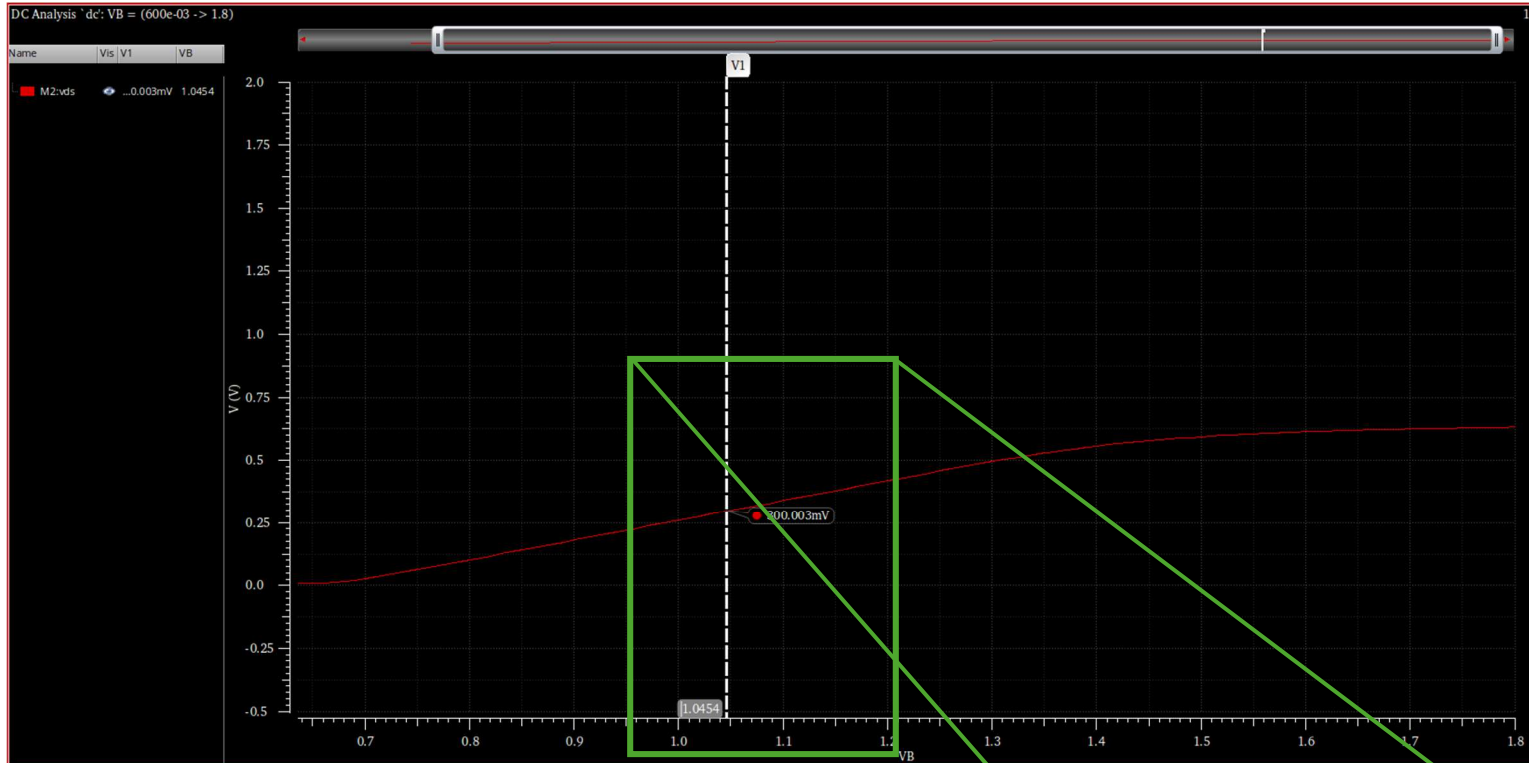
Schematic



VB

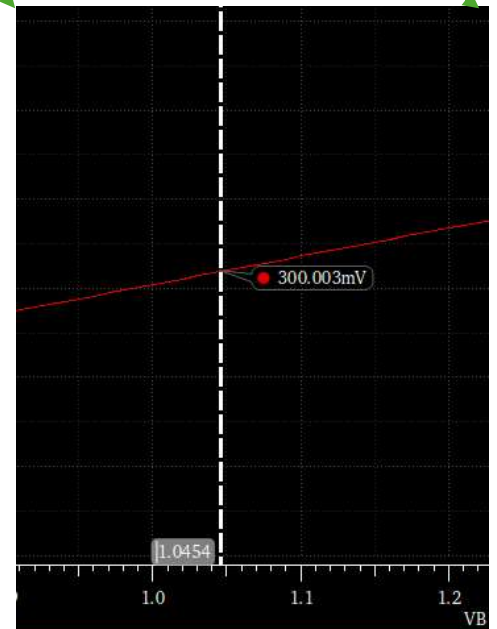
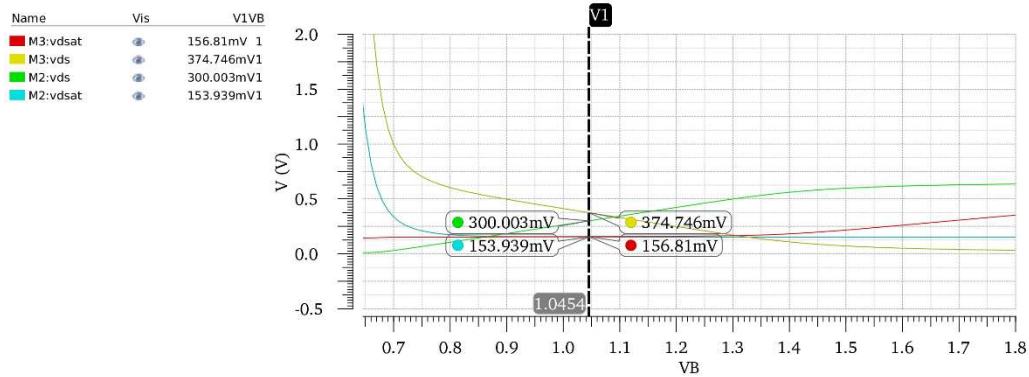
We want to choose a suitable V_B to keep both transistors in saturation and allow the bias current forced by current source to path so we will do a sweep and we won't start the sweep from zero to avoid simulation errors or crashes as at $V_B=0$ no current will pass but the current source will still try to force current of V_B and will set we will take the point at which V_{ds} M2 is $V^*+100mv=300mv$

VDS M2 vs VB



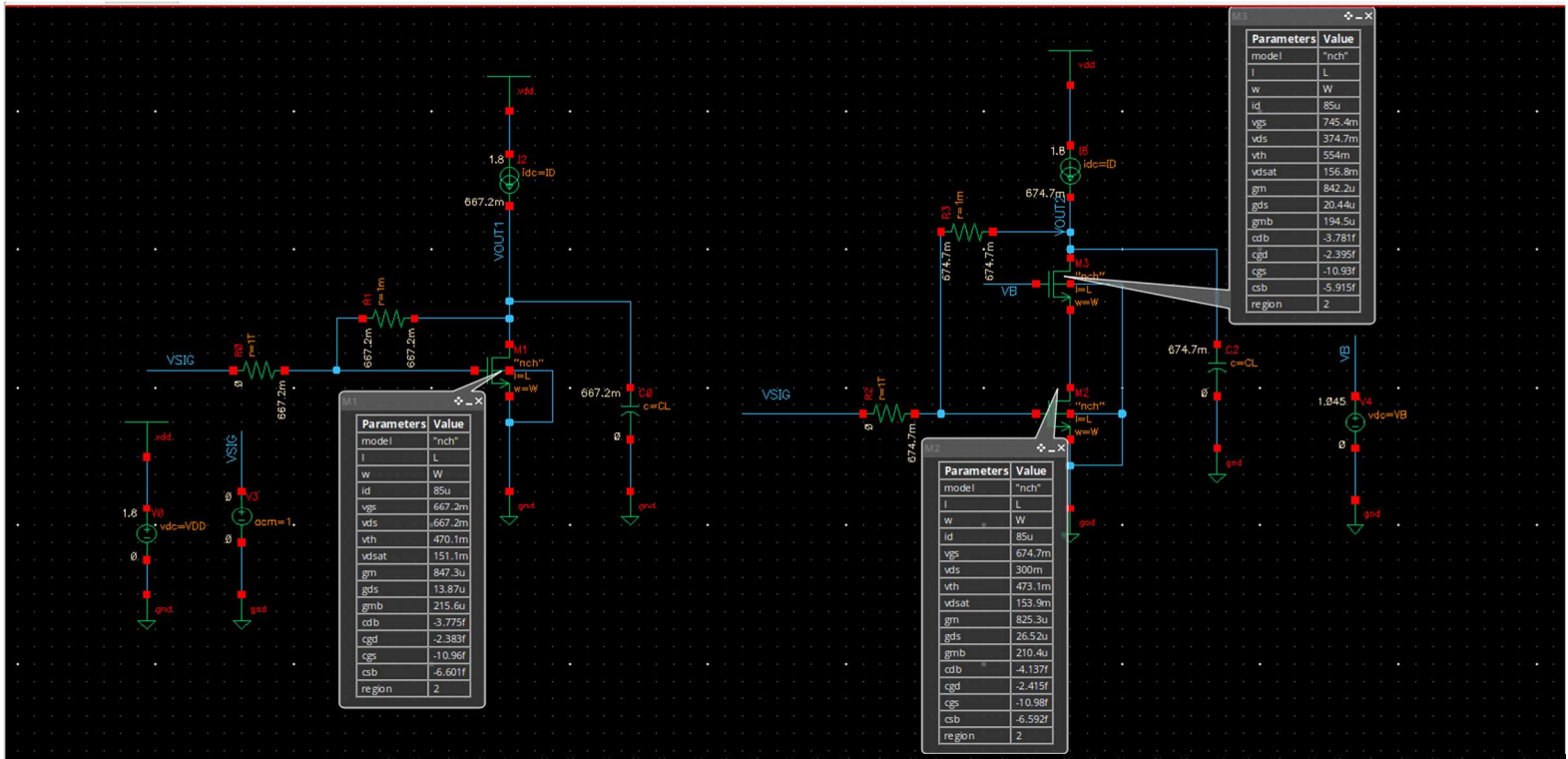
So VB op= 1.0454v which ensures both mosfets are in saturation

DC Analysis `dc`: VB = (600e-03 -> 1.8)

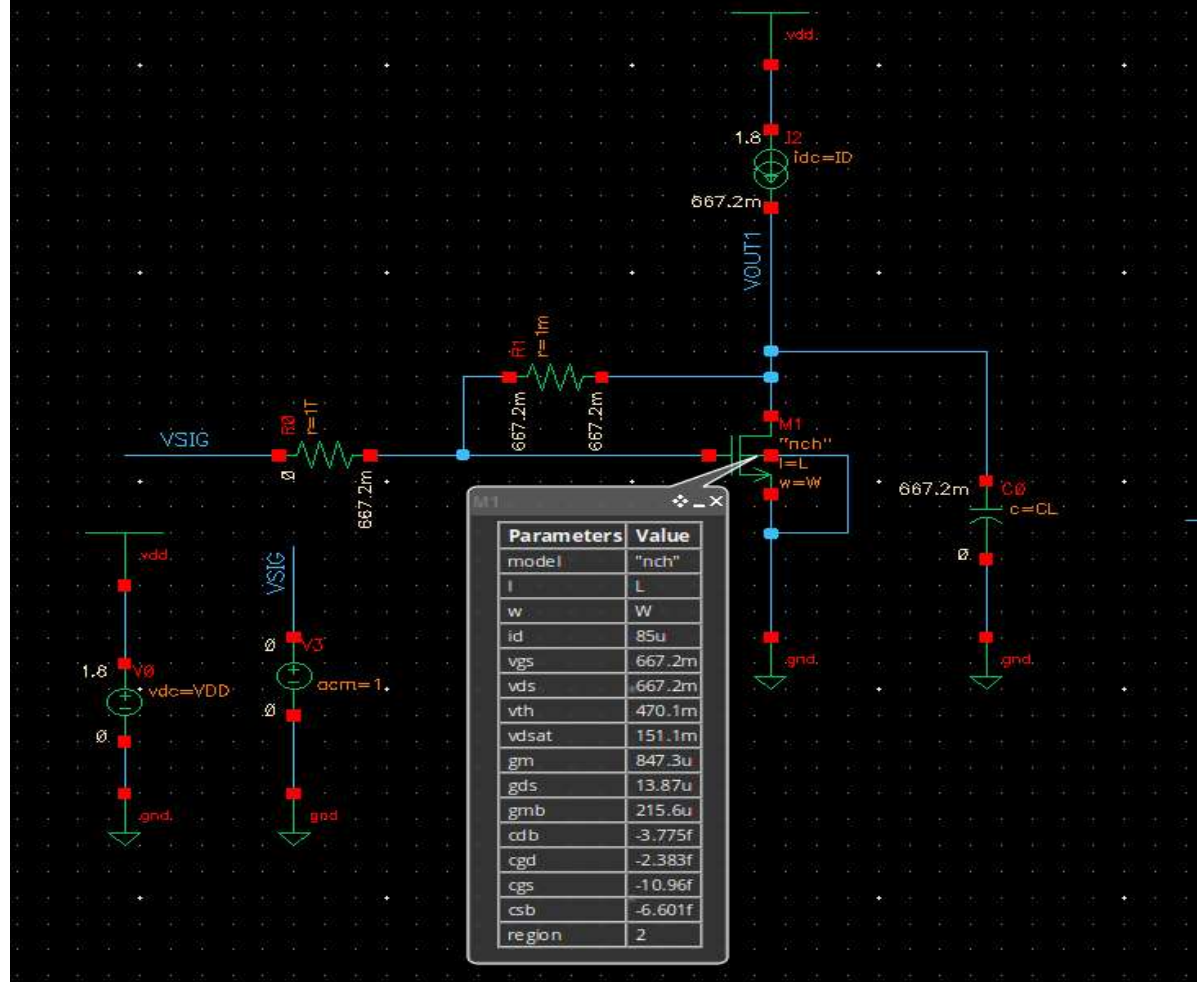


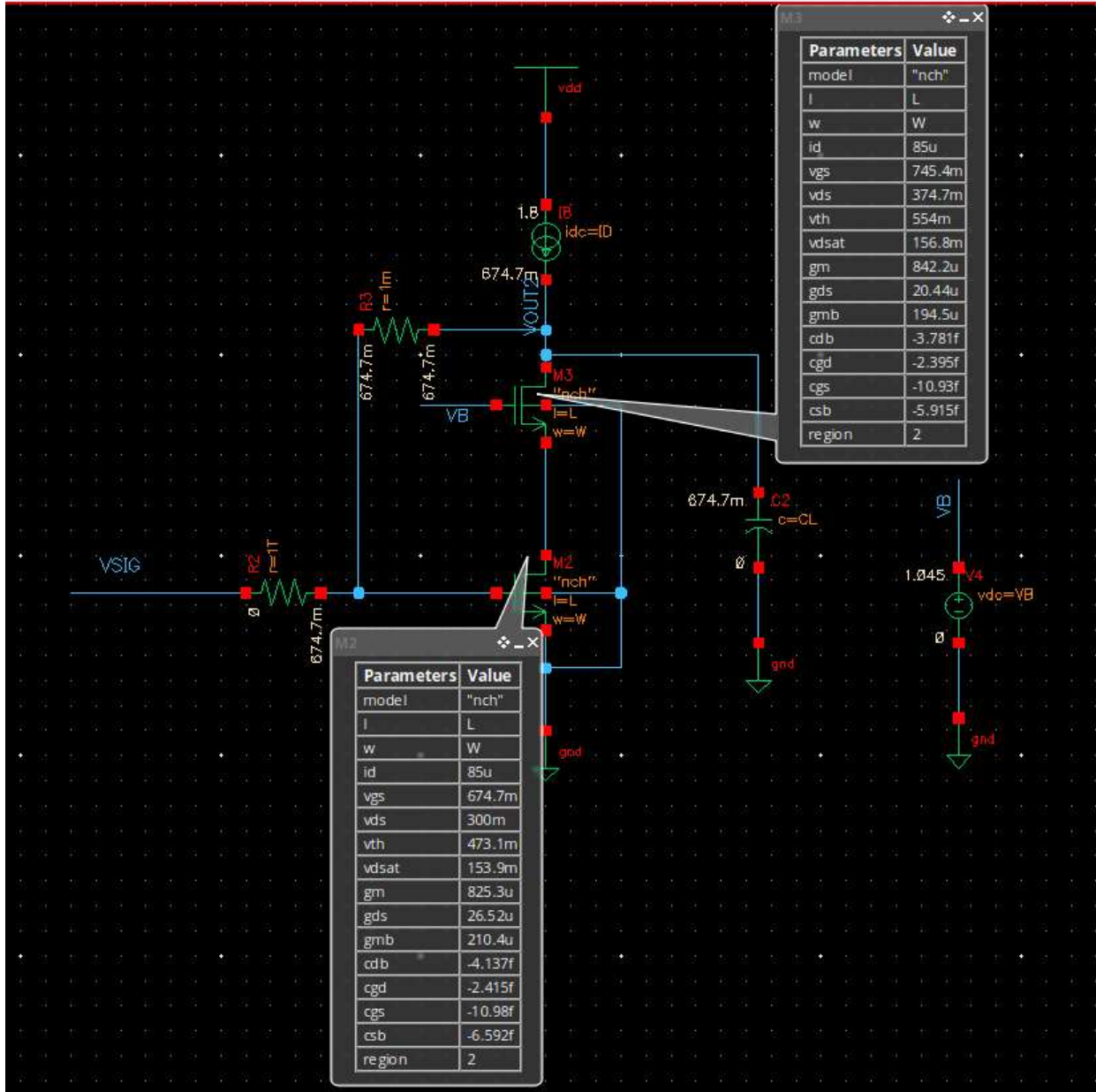
Here is a magnified image of VB op

Schematic with op and node voltages annotated



Common source AMP
op for clearer view





Cascoded AMP op for clearer view

Feedback Resistance settings

Edit Object Properties

Browse Reset Instance Labels Display

Property Value Display

Library Name analogLib off

Cell Name res off

View Name symbol off

Instance Name R0 off

Add Delete Modify

CDF Parameter Value Display

Model name off

Resistance 1T Ohms off

Length off

Width off

Multiplier off

Scale factor off

Temp rise from ambient off

Temperature coefficient 1 off

Temperature coefficient 2 off

Resistance Form off

Generate noise? off

Capacitance off

Alias for Lin. temp. co. off

Alias for Quad temp. co. off

Lin temp co of lin cap off

Quad temp co of lin cap off

Resistance Scaling Factor off

Capacitance Scaling Factor off

AC resistance 1m Ohms off

Temperature difference off

Input Resistance Settings

They are set this way so we can trick the simulator, so it biases the mosfet with the correct values for saturation as a diode connected and use also as an amplifier

Edit Object Properties

Browse Reset Instance Labels Display

Property Value Display

Library Name analogLib off

Cell Name res off

View Name symbol off

Instance Name R1 off

Add Delete Modify

CDF Parameter Value Display

Model name off

Resistance 1m Ohms off

Length off

Width off

Multiplier off

Scale factor off

Temp rise from ambient off

Temperature coefficient 1 off

Temperature coefficient 2 off

Resistance Form off

Generate noise? off

Capacitance off

Alias for Lin. temp. co. off

Alias for Quad temp. co. off

Lin temp co of lin cap off

Quad temp co of lin cap off

Resistance Scaling Factor off

Capacitance Scaling Factor off

AC resistance 1T Ohms off

Temperature difference off

5) check for saturation

As we saw in the ballon all mosfets are in region 2 which is saturation .

6) Vth for mosfets

No ,not all mosfets here have the same Vth and we can see this clearly in mosfet 3 as its Vth is higher than the rest and this because of the body effect and the depletion region created by the N and P junction in channel between the body and the two N region as VSB increase the depletion increase the more voltage by gate needed to create the channel and this appears clearly as mosfet 3 body terminal is not connected to its source .

M1 Vth=470.1m
M2 Vth=473.1m
M3 Vth=554m

7) gm>>gds

As $g_{ds} = \frac{\partial I_D}{\partial V_{DS}}$, $gm = \frac{\partial I_D}{\partial V_{GS}}$ and Id changes aggressively with Vgs according to the square law which gives a large gm while of Id Vds is doesn't that much in saturation region resulting in a large ro and small gds .

8) gm>gmb

Though it can be sometimes approximated gm>>gmb and we neglect gmb as $gm = \frac{\partial I_D}{\partial V_{GS}}$, $g_{mb} = \frac{\partial I_D}{\partial V_{BS}}$ and the change of Id with Vbs is as strong as the change of Id with Vgs.

9)Cgs > Cgd

Because in saturation pinch off occurs and due to it the charge near the drain are less than that near source.

10) Csb > Cdb

As the drain and source act as reverse biased diodes with bulk and the drain voltage is higher than source so the depletion region width is greater at drain than source so the capacitance is greater at source as it is inversely proportional to width .

AC Analysis

1) Setup

Choosing Analyses -- ADE L (1)

Analysis: ☐ tran ☐ dc ☒ ac ☐ noise
☐ xf ☐ sens ☐ dmatch ☐ acmatch
☐ stb ☐ pz ☐ sp ☐ envlp
☐ pss ☐ pac ☐ pstb ☐ pnoise
☐ pxf ☐ psp ☐ qpss ☐ qpac
☐ qpnoise ☐ qpxf ☐ qpssp ☐ hb
☐ hbac ☐ hbnoise ☐ hbbsp

AC Analysis

Sweep Variable: ☒ Frequency
☐ Design Variable
☐ Temperature
☐ Component Parameter
☐ Model Parameter
☐ None

Sweep Range: ☒ Start-Stop Start: 1 Stop: 10G
☐ Center-Span

Sweep Type: ☒ Points Per Decade 10
☒ Logarithmic ☐ Number of Steps

Add Specific Points: ☐

Specialized Analyses: None

Enabled: ☒

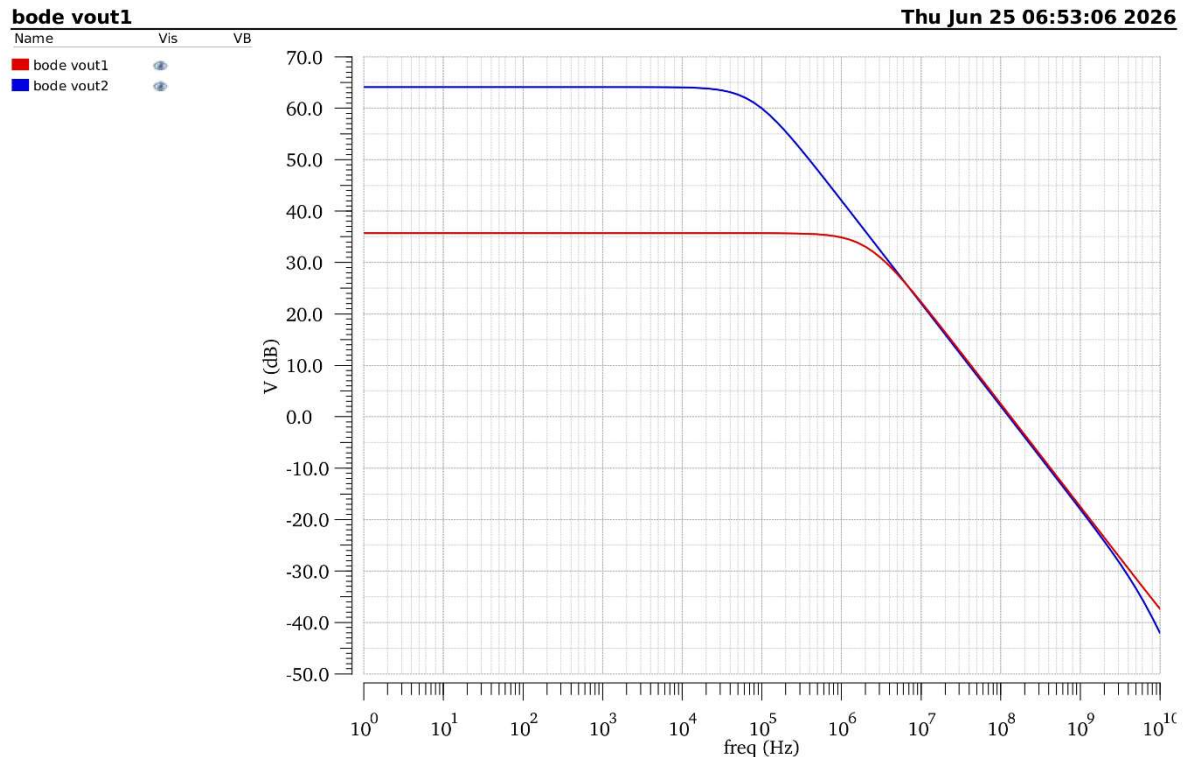
Options...

OK Cancel Defaults Apply Help

2) DC gain, BW, GBW, and UGF

Test	Output	Nominal	Spec	Weight	Pass/Fail
lab3:ac-analysis:1	db DC gain vout 1	35.72			
lab3:ac-analysis:1	DC gain vout1	61.09			
lab3:ac-analysis:1	BW vout1	2.196M			
lab3:ac-analysis:1	GBW vout1	134.5M			
lab3:ac-analysis:1	UGF vout1	134.7M			
lab3:ac-analysis:1	db DC gain vout 2	64.13			
lab3:ac-analysis:1	DC gain vout2	1.61k			
lab3:ac-analysis:1	BW vout2	78.93k			
lab3:ac-analysis:1	GBW vout2	127.3M			
lab3:ac-analysis:1	UGF vout2	127.4M			

3) Bode plot of CS and cascode appended



It's obvious that the BW of the CS is higher than the cascoded amplifier as the GBW of both is close but the gain of the cascaded amplifier is much higher than that of the CS amplifier and also we can think about that $BW \approx \frac{1}{2\pi R_{out} C_{out}}$ and the C_{out} is relatively the same in both of the cascoded and CS Amplifier but the R_{out} is much higher in the cascoded amplifier resulting in a lower BW.

4) Hand analysis

Common Source

Given:

$$g_m = 847.3 \mu S, \quad g_{ds} = 13.87 \mu S, \quad C_L = 1 \text{ pF}$$

Using Dominant-pole approximation as we neglect the parasitic capacitance as $C_L \gg C_{gs} + C_{gd} + C_{DB} + C_{SB}$

$$R_{out} \simeq r_o = \frac{1}{g_{ds}} = \frac{1}{13.87 \times 10^{-6}} \Omega$$

$$Z_{out}(s) = r_o \parallel \frac{1}{sC_L} = \frac{r_o}{1 + sr_oC_L}$$

$$A_v(s) = \frac{V_{out}}{V_{in}} = -g_m Z_{out}(s) = \frac{-g_m r_o}{1 + sr_oC_L}$$

$$\therefore \omega_p = \frac{1}{r_o C_L}, \quad f_c = BW = \frac{1}{2\pi r_o C_L}$$

1) BW

$$BW = \frac{1}{2\pi R_{out} C_L} \simeq \frac{1}{2\pi r_o C_L}$$

$$BW = \frac{1}{2(72098)(10^{-12})}$$

$$BW = 2.207 \times 10^6 \text{ Hz}$$

$BW = 2.207 \text{ MHz}$

2) DC gain

$$A_{v0} = g_m r_o = g_m \left(\frac{1}{g_{ds}} \right)$$

$$A_{v0} = 847.3 \times 10^{-6} \left(\frac{1}{13.87 \times 10^{-6}} \right) \approx 61$$

$$A_{v0, \text{dB}} = 20 \log_{10}(61) = 35.7 \text{ dB}$$

$A_{v0} = 61, \quad A_{v0, \text{dB}} = 35.7 \text{ dB}$
--

3) GBW

$$GBW = A_{v0} \times BW = 61 \times 2.207 \times 10^6$$

$$GBW = 134.627 \times 10^6 \text{ Hz}$$

$$GBW = 134.627 \text{ MHz}$$

4) UGF

$$|A_v(f)| = \frac{A_{v0}}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

$$1 = \frac{61}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

$$\left(\frac{f}{f_c}\right)^2 = 61^2 - 1 = 3720$$

$$\frac{f}{f_c} = \sqrt{3720} \approx 61$$

$$\therefore f = f_c \sqrt{61^2 - 1} \approx BW \times 61 \approx GBW$$

$$UGF \approx 134.6 \text{ MHz}$$

Cascode

Given from OP:

$$M_2: g_{m2} = 825.3 \mu\text{S}, \quad g_{ds} = 26.52 \mu\text{S}, \quad r_{o2} = 37.707 \text{ k}\Omega$$

$$M_3: g_{m3} = 842.2 \mu\text{S}, \quad g_{mb} = 194.5 \mu\text{S}, \quad g_{ds3} = 20.44 \mu\text{S}, \quad r_{o3} = 48.924 \text{ k}\Omega$$

$$C_L = 1 \text{ pF}, \quad g_{m3} + g_{mb3} = 1036.7 \mu\text{S}$$

Dominant-pole approximation:

$C_L \gg$ parasitic capacitors $\Rightarrow$ output pole dominates

$$R_{out} = r_{o2}[1 + (g_{m3} + g_{mb})r_{o3}] + r_{o3}$$

$$R_{out} = 37.707 \times 10^3 [1 + (1036.7 \times 10^{-6})(48.924 \times 10^3)] + 48.924 \times 10^3$$

$$R_{out} \simeq 1.9719 \times 10^6 \Omega = 1.9719 \text{ M}\Omega$$

$$A_v(s) \simeq \frac{-g_{m2}R_{out}}{1 + sR_{out}C_L}$$

$$\therefore \omega_p = \frac{1}{R_{out}C_L}, \quad f_c = BW = \frac{1}{2\pi R_{out}C_L}$$

$$R_{out} = 1.9719 \text{ M}\Omega$$

1) BW

$$BW = \frac{1}{2\pi R_{out}C_L}$$

$$BW = \frac{1}{2\pi(1.9719 \times 10^6)(10^{-12})}$$

$$BW = 80.7 \times 10^3 \text{ Hz}$$

$$BW = 80.7 \text{ kHz}$$

2) DC gain

$$A_{v0} = -g_{m2}R_{out} \Rightarrow |A_{v0}| = g_{m2}R_{out}$$

$$|A_{v0}| = (825.3 \times 10^{-6})(1.97198 \times 10^6) \approx 1626.8$$

$$A_{v0,\text{dB}} = 20\log_{10}(1626.9) = 64 \text{ dB}$$

$$|A_{v0}| = 1626.8, \quad A_{v0,\text{dB}} = 64 \text{ dB}$$

3) GBW

$$GBW = |A_{v0}| \times BW$$

$$GBW = 1626.8 \times 80.7 \times 10^3$$

$$GBW = 131.3 \times 10^6 \text{ Hz}$$

$$GBW = 131.3 \text{ MHz}$$

4) UGF

$$|A_v(f)| = \frac{|A_{v0}|}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

$$1 = \frac{1626.8}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

$$\left(\frac{f}{f_c}\right)^2 = 1626.8^2 - 1$$

$$\frac{f}{f_c} = \sqrt{1626.8^2 - 1} \approx 1626.8$$

$$\therefore f = f_c \sqrt{1649.9^2 - 1} \approx BW \times 1626.8 \approx GBW$$

$$UGF \approx 131.3 \text{ MHz}$$

5) Hand analysis vs simulation in a Table

Circuit	Parameter	Hand Analysis	Simulation	Difference
Common Source	DC Gain	61 V/V	61.09 V/V	$+0.15$
Common Source	DC Gain in dB	35.70 dB	35.72 dB	$+0.02\text{ dB}$
Common Source	BW	2.207 MHz	2.196 MHz	-0.50
Common Source	GBW	134.627 MHz	134.5 MHz	-0.09
Common Source	UGF	134.6 MHz	134.7 MHz	$+0.07\%$
Cascode	DC Gain	1.626 kV/V	1.61 kV/V	-2.42%
Cascode	DC Gain in dB	64 dB	64.13 dB	-0.22 dB
Cascode	BW	80.7 kHz	78.93 kHz	-2.24%
Cascode	GBW	131.3 MHz	127.3 MHz	-3.08%
Cascode	UGF	131.3 MHz	127.4 MHz	-3.01%

6) Comment

The simulation results are pretty much close to the hand analysis for both circuits. The small differences are due to neglected parasitic capacitances, body effect, and the dominant-pole approximation used in hand analysis. The cascode amplifier has much higher DC gain than the common-source amplifier because its output resistance is larger. However, this larger output resistance causes a reduction in the bandwidth due to the larger ($R_{out}C_L$) time constant. The GBW and UGF of both circuits are close, showing a gain-bandwidth trade-off.

Appendix

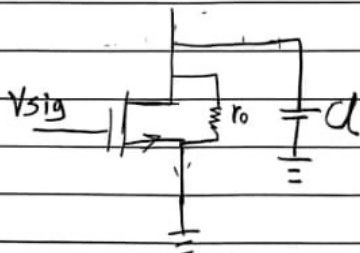
Hand analysis

Date: _____

Subject: _____

Hand Analysis

Common Source



We can use approximation of the Dominant pole and neglect the parasitic capacitors as $CL \gg Cgs, Cgd, CDB, CSB$

$$BW = \frac{1}{R_{out} C_L}$$

$$\text{as } \frac{V_{out}}{V_{in}} = -g_m Z_{out}$$

$$-g_m \frac{r_o}{sC_L r_o + 1}$$

so the -3db pole is at $\frac{1}{R_{out} C_L}$

$$1) BW = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi 10^{-12} \times \frac{1}{13.87 \times 10^{-6}}} = \boxed{2.207 \text{ MHz}}$$

$$2) |DC \text{ gain}| = g_m r_o = 847.3 \times 10^{-6} \times \frac{1}{13.87 \times 10^{-6}} = \boxed{61}$$
$$20 \log 61 = \boxed{35.7 \text{ dB}}$$

1

Date: _____

Subject: _____

3) GBW

$$\text{gain} \times \text{BW} = 61 \times 2.207 \times 10^6 = \boxed{134.627 \text{ MHz}}$$

4) UGF

$$\frac{-g_m r_o}{s r_o C + 1}$$

$$\frac{g_m r_o}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}} = 1, \quad 61^2 = 1 + \left(\frac{f}{f_c}\right)^2$$

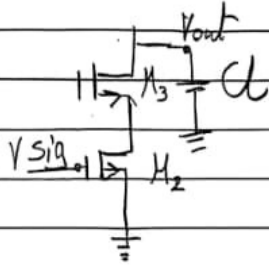
$$\sqrt{3720} = \frac{f}{f_c}, \quad f = f_c \times 61 = \text{BW} \times 61 = \text{GBW}$$

$$\text{UGF} = \text{GBW} = \boxed{134.6 \text{ MHz}}$$

Date: _____

Subject: _____

CasCode



$$R_{out} = r_{o2} (1 + (g_{m3} + g_{mb3}) r_{o3}) + r_{o3} = 51 \times 37707 + 48923 = 1971980 \Omega$$

$$1) g_m = (g_{m2}) = 825.3 \times 10^{-6}$$

$$\text{DC gain } g_m R_{out} = 1626.8$$

$$\text{in dB} = 20 \log 1626.8 = 64 \text{ dB}$$

$$2) BW = \frac{1}{2\pi \times 1971980 \times 10^{-12}} = 8070.8 \text{ Hz}$$

$$3) GBW = 8070.8 \times 1626.81 = 131.3 \text{ MHz}$$

$$4) UGF = GBW = 131.3 \text{ MHz}$$